

Homework 8

due 06.12.2023, 14:00

Exercise 8.1 - Cartesian Communicators (10 points)

Use MPI to set up a cartesian communicator for a 4×12 process grid.

- Read the 400×600 matrix from file
`/work/korzec/LAB2/ex8/notstirred.txt`
on stromboli, stored in the same format as in the two previous exercises.
Process 0 is responsible for all I/O and only local arrays of size 100×50 are allowed.
- Process 0 should read the contents of the file `shifts.dat` and broadcast it. The file contains 1000 rows with two integers in each. Depending on the values of those two integers, different operations on the matrix are performed. If the values are j and k , we distinguish the cases:
 - $j == 0$: ranks with coordinates $(:, k)$ perform a circular shift among themselves where (i, k) sends its local 100×50 matrix to $(i + 1 \bmod 4, k)$.
 - $j == 1$: ranks with coordinates $(:, k)$ perform a circular shift among themselves where (i, k) sends to $(i + 3 \bmod 4, k)$.
 - $j == 2$: ranks with coordinates $(k, :)$ perform a circular shift among themselves where (k, i) sends to $(k, i + 1 \bmod 12)$.
 - $j == 3$: ranks with coordinates $(k, :)$ perform a circular shift among themselves where (k, i) sends to $(k, i + 11 \bmod 12)$.
- Finally process 0 collects and prints the transformed matrix in the usual file format. Run `check.m` on both original and final matrix and attach the resulting images to your PDF (adjust the file-name in the script). You can also use the script to visualize intermediate results for debugging.